E9 241 Digital Image Processing Practice Problem Set 3

1. **Orthogonality of Basis Functions**
   Let the Discrete Fourier transform (DFT) of an $M \times N$ image $f(m,n)$ be $F(u,v)$, where the inverse formula is given by

$$f(m,n) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) \exp\left[jmu\frac{2\pi}{M} + jnv\frac{2\pi}{N}\right].$$

   Prove that the basis functions in the inverse formula are orthogonal to each other i.e.

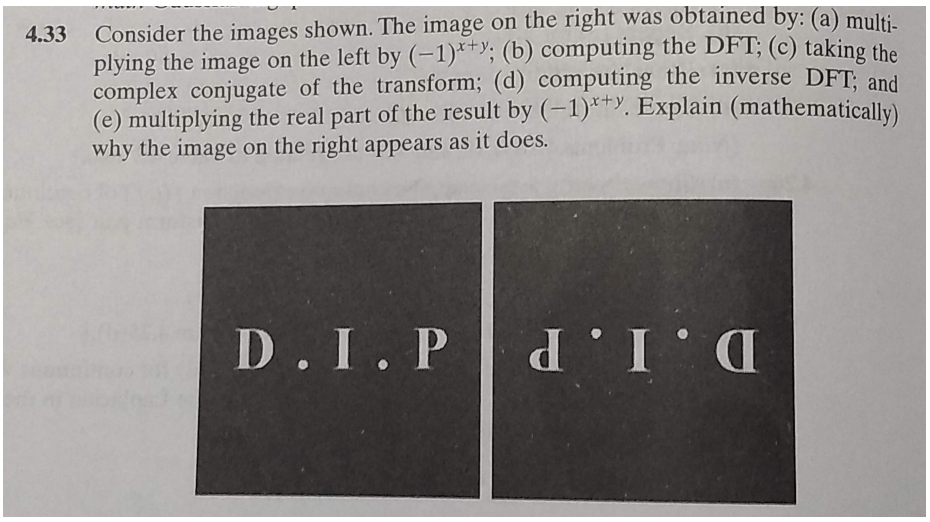
$$\frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \exp\left[jmu_1\frac{2\pi}{M} + jnv_1\frac{2\pi}{N}\right] \exp\left[-jmu_2\frac{2\pi}{M} - jnv_2\frac{2\pi}{N}\right] = \delta(u_1 - u_2, v_1 - v_2),$$

   where $u_1, u_2 \in \{0, 1, 2, \ldots, M_1\}$ and $v_1, v_2 \in \{0, 1, 2, \ldots, N-1\}$.

2. **Parseval's Theorem**
   Prove that Parseval's theorem holds for the DFT assuming that $f(m,n)$ is a real, i.e.

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} (f(m,n))^2 = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} |F(u,v)|^2.$$

3. **Geometric transform using DFT**
   Problem 4.33 from Gonzalez and Woods



4.33 Consider the images shown. The image on the right was obtained by: (a) multiplying the image on the left by $(-1)^{x+y}$; (b) computing the DFT; (c) taking the complex conjugate of the transform; (d) computing the inverse DFT; and (e) multiplying the real part of the result by $(-1)^{x+y}$. Explain (mathematically) why the image on the right appears as it does.

4. **Equivalent filter**
Problem 4.27 from Gonzalez and Woods

★**4.27** Consider a 3×3 spatial mask that averages the four closest neighbors of a point (x, y), but excludes the point itself from the average.

(a) Find the equivalent filter, $H(u, v)$, in the frequency domain.

(b) Show that your result is a lowpass filter.